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June 25, 1956

MEMORANDUM TO: Mr. J. A. Haddad

SUBJECT: Meeting on "Next Generation of Computers," June 15, 1956

LOCATION: Poughkeepsie Research Laboratory

IN ATTENDANCE:

Dr. W. Buchholz	J. Haddad
J. Coombs	Dr. C. Hurd
S. Dunwell	W. Lawless
P. Fagg	D. Pendery
R. Gregory	R. Rabenkamp
J. Griffith	N. Rochester

PURPOSE: The purpose of the meeting was to enable engineering representatives to present their thinking on the points raised in Dr. Hurd's memorandum of April 19th: "Next Generation of Computers."

SUMMARY:

1. Engineering believes that the binary-decimal question should no longer be considered of major importance since future machines will be able to operate in at least both of these modes.

2. Because we must assume occasional machine errors, future equipment must incorporate systems of error detection and correction adequate to provide a predictable high degree of accuracy.

3. The input-output system being planned for the STRETCH project will provide adequate generality for future machines.

The Nature of Character Representation

The subject of character representation; that is binary, decimal, or some combination of the two, was discussed at some length. With respect to the arithmetic and logical units of future machines, Engineering plans to provide necessary facilities to permit at least binary and decimal arithmetic, alphanumeric logical operations, and single bit logical operations. This concept may well be extended to permit the handling of any modulus up to, say, 64 on more powerful machines.

With respect to memory addressing, it was generally agreed that a binary number system must be used. However, because of the general practice of writing programs in symbolic notation and preparing cross reference lists between symbolic addresses and actual addresses, Engineering believes that binary memory addressing will not present serious operating problems. However, if it develops that this is not a valid assumption, Engineering will provide automatic decimal-binary conversion, or the equivalent, for address notation.

It was generally agreed that it is important from a sales point of view to be able to say that future machines can be operated on a decimal basis, without knowledge of binary arithmetic; however, it is anticipated that as users become generally more familiar with the computing art they will want to understand the logic of their machines fully in order to make maximum use of the binary, decimal, etc., capacity of future machines.

In general, Engineering believes that the binary-decimal question should no longer be considered of major importance since future machines will be able to operate in at least both of these modes.

The Extent of Checking, Error Detection and Fault Detection

These three subjects, plus Error Correction and Fault Correction were defined and discussed. The Engineering research activities in the area of information research and systems of coding which permit automatic error correction were presented. Some of the different systems' requirements with respect to machine performance were also discussed; e.g. military products, high speed scientific computing, in-line commercial processing. Engineering generally believes that in order to attain the maximum effectiveness from our components we must assume occasional machine errors. Therefore, it is extremely important that systems of error detection and automatic correction be developed to a high degree. Our goal is to develop equipment which will process jobs with a predictable high degree of accuracy, even under conditions in which the equipment is not working perfectly. An equally important objective is to develop equipment in which fault detection and correction are simplified to the highest degree possible.

Generality of Input-Output Equipment

Engineering research into systems for generalizing input-output equipment facilities, as represented by the input-out system plan for the STRETCH

system, was presented. We generally agreed that if the STRETCH input-output system or its equivalent materializes in future equipment, adequate generality will have been provided.

W. J. Lawless, Jr.

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